RAISE : Representation Agnostic Information and Separation Estimation

Anonymous Author(s)
Affiliation
Address
email

Abstract

Waveform sensors that resolve sub-footprint structure from overlapping echoes fail in ways usually attributed to architecture or training. We show this failure is instead an information limit: below a sensor-dependent separation regime, the information available for estimating return separation collapses, imposing a representation-independent performance ceiling that no representation, learned or otherwise, can exceed. We prove this as a representation-agnostic degradation bound, composing a marginal Fisher-information collapse with a non-asymptotic mutual-information result and the data-processing inequality, and confirm it on real GEDI spaceborne lidar data. We then operationalize the same limit into the Ziv-Zakai Headroom Estimator (ZZHE), a diagnostic that separates fixable from fundamental error for a deployed estimator, and validate it against independently-sourced airborne LiDAR ground truth, where all tested separation bins pass a conservative falsifiability margin. This establishes where estimator improvement remains possible and where the sensor's information limit becomes unavoidable.

1 Introduction

Consider a waveform sensor observing two reflecting surfaces whose separation is small relative to the width of its own pulse. As that separation shrinks, the information available to recover it from the observation collapses, and no amount of averaging, denoising, or architectural sophistication can recover more than what remains: the two returns increasingly overlap into a signal from which the separation becomes progressively harder, and eventually practically impossible, to distinguish. This limitation is not primarily attributable to the estimator itself, but rather reflects an intrinsic constraint imposed by the observation.

GEDI, the spaceborne waveform lidar used to estimate global forest canopy height, shows exactly this failure mode: waveform broadening from transmit pulse width and terrain slope merges canopy and ground returns (Cai et al., 2025), a pattern already documented in large-footprint waveform lidar before GEDI itself launched (Hancock et al., 2012), and compounded by a residual footprint-level geolocation uncertainty of roughly ten metres in the mission's own processing (Beck et al., 2021). In every case the degradation is explained after the fact, never predicted from sensor physics in advance.

The natural response, in both remote sensing and representation learning, is to treat this as fixable by better modeling. Four bodies of work sit closest to that hope. Identifiability theory gives both sides of it: unsupervised disentanglement is provably unidentifiable without inductive bias (Locatello et al., 2019), yet nonlinear ICA becomes identifiable once an auxiliary conditioning variable is available (Khemakhem et al., 2020). Our answer sits outside that trade: it is not about what structure could rescue identifiability, but about a physical information floor that no structure changes. The bound itself composes a non-asymptotic Fisher-to-mutual-information result (Górecki et al., 2024) with the classical data-processing inequality for Fisher information (Zamir, 1998), turning a local,

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

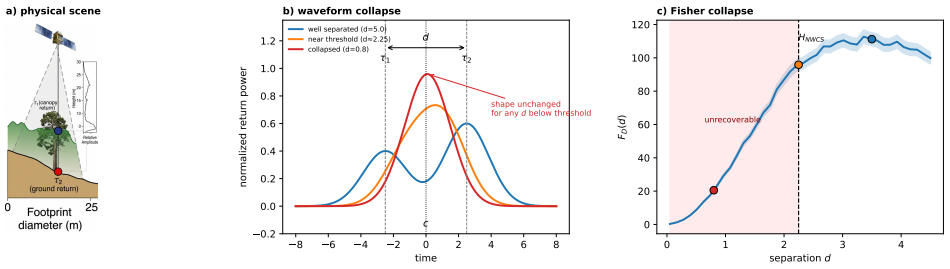


Figure 1: The identifiability collapse. (a) A waveform footprint with two overlapping returns. (b) The observed waveform for three separations; below threshold, shape no longer changes with d. (c) Marginal Fisher information $F_D(d)$, vanishing in the shaded region below H_{NWCS}.

pointwise estimator statement into a global, representation-agnostic one. Our diagnostic draws on an older lineage, the Ziv-Zakai bound, a detection-theoretic alternative to the Cramér-Rao bound for problems where error does not vanish smoothly (Ziv and Zakai, 1969), later made explicit for a resolution-limited problem structurally close to ours (Bell et al., 1996). Coordinate-based implicit representations (Tancik et al., 2020) and Earth observation foundation models such as SatMAE (Cong et al., 2022) and Prithvi (Jakubik et al., 2023) are evaluated against this same degradation empirically, but none ask, before training, whether the information was ever present to begin with. Our primary contributions involve proving a representation-agnostic degradation bound that establishes a fundamental performance ceiling for resolving overlapping echoes, confirmed via real GEDI spaceborne lidar data. We also operationalize this bound into the Ziv-Zakai Headroom Estimator (ZZHE) a diagnostic tool separating fixable from fundamental errors and validate it against independent airborne LiDAR ground truth.

2 Theory

Consider a waveform sensor observing two overlapping returns from a single footprint,

$$W_{\mathrm{obs}}(t) = W_{\mathrm{pred}}(t; \tau_1, \tau_2) + N(t), \qquad N(t) \sim \mathcal{N}(0, \sigma_n^2), \tag{1}$$

where

$$W_{\mathrm{pred}}(t; \tau_1, \tau_2) = w_1 P(t, \tau_1, \sigma_{\mathrm{pulse}}) + w_2 P(t, \tau_2, \sigma_{\mathrm{pulse}}). \tag{2}$$

We reparameterize the two arrival times as a centroid and a separation,

$$c_{\mathrm{ctr}} = \frac{\tau_1 + \tau_2}{2}, \qquad d = \tau_2 - \tau_1, \tag{3}$$

so that d carries the structure we want to recover and c_{ctr} is a nuisance describing where that structure sits. (We use the subscript throughout to keep this centroid distinct from the speed of light, which appears with its own subscript in Section 3.) Figure 1 shows this model directly: a real overlapping-echo footprint (a), the resulting waveform collapsing as d shrinks (b), and the marginal Fisher information $F_D(d)$ vanishing below the threshold H_{NWCS} (c).

Since c_{ctr} is never given to us, the honest quantity is not the Fisher information at some fitted value of c_{ctr}, but what remains once c_{ctr} has been integrated out,

$$p(x \mid d) = \int p(x \mid c_{\mathrm{ctr}}, d)\, p(c_{\mathrm{ctr}})\, dc_{\mathrm{ctr}}, \qquad F_D(d) = \mathbb{E}_{X|d}\left[\left(\frac{\partial}{\partial d} \log p(X \mid d)\right)^2\right]. \tag{4}$$

A profile-likelihood or Schur-complement treatment would instead condition on a single fitted c, discarding exactly the uncertainty that makes this problem hard; $F_D(d)$ keeps it. As the two returns merge, the Fisher information of $\theta = (\tau_1, \tau_2)$,

$$\det I(\theta) = \frac{w_1^2 w_2^2}{\sigma_n^4}\left[R_{P'P'}(0)^2 - R_{P'P'}(\tau_1 - \tau_2)^2\right], \tag{5}$$

with $R_{P'P'}$ the autocorrelation of the pulse derivative, vanishes: their derivative waveforms become indistinguishable, and the observation becomes locally blind to d. For an unbiased estimator, this

Table 1: Real GEDI validation summary.

Metric	Value	Rules in / rules out
Roughness corr. (vs. Copernicus DEM)	$r = 0.612$, $p < 0.001$, N=7,894	structural signal, not noise
Geolocation corr. (vs. Copernicus DEM)	$r = 1.000$ (-16 m datum offset)	rules out gross geolocation error
Off-nadir incidence correction (max)	$\leq 0.139\%$	rules out incidence-angle artifact
Error reduction, below $\rightarrow$ above threshold	3–6$\times$	matches predicted threshold transition

shows up as an unbounded Cramér–Rao variance near collapse. But an unbiased estimator is a narrow target, and nothing so far rules out a biased or learned estimator quietly doing better.

Let $X = W_{\mathrm{obs}}$, and let any downstream representation, learned or otherwise, be written as $Y = f(X)$. Because representation happens strictly after sensing, the data-processing inequality gives $I(Y; D_{\mathrm{sep}}) \leq I(X; D_{\mathrm{sep}})$, which already says no representation can see more than the waveform itself contains. A non-asymptotic result of Górecki et al. (2024) turns the marginal Fisher information above into exactly such a ceiling,

$$I(X; D_{\mathrm{sep}}) \leq \ln\left(1 + \frac{1}{2}\int_0^D \sqrt{F_D(d)}\, dd\right). \tag{6}$$

Theorem (Representation-Agnostic Degradation Bound)**.** *For the two-echo waveform model and any deterministic representation $Y = f(W_{\mathrm{obs}})$,*

$$I(Y; D_{\mathrm{sep}}) \leq I(X; D_{\mathrm{sep}}) \leq \ln\left(1 + \frac{1}{2}\int_0^D \sqrt{F_D(d)}\, dd\right). \tag{7}$$

As this bound approaches zero near collapse, the information available to every downstream representation approaches zero with it, and by Fano's inequality, no estimator built on Y can reliably discriminate between sufficiently close separations.

The ceiling is representation-independent by construction: a learned representation can recover information the waveform actually contains, but it cannot manufacture information the sensing process has already destroyed. For the sensor configuration studied here, the bound evaluates to $I(X; D_{\mathrm{sep}}) \leq 2.00$ nats, independent of the threshold prefactor κ disclosed in Section 3.

3 Results

Real GEDI validation. Applied to real GEDI L2B data over Colorado, the threshold predicts a transition matching observed estimator behavior: a neural-field estimator and an inverse-distance weighted baseline both show a three- to six-fold reduction in error moving from below to above threshold, and this transition survives synthetic grid-convergence checks and the confound controls in Table 1.[1]

From ceiling to diagnostic. Theorem 2 establishes that a threshold exists; it does not tell a practitioner, for a specific deployed estimator, how much of its current error is fixable. We convert the same marginal Fisher information into a usable number through the Ziv-Zakai bound, adapted here to a centroid-marginalized detection problem (Bell et al., 1996), under a pointwise (test-point) construction rather than the classical global-prior form of the bound; the exact scope of this construction, including the local-homogeneity assumption it relies on, is stated in full in Appendix F. Given a deployed estimator's empirical error $L_{\mathrm{current}}(d)$, Algorithm computes a theoretically justified, estimator-agnostic floor $L_{\mathrm{ZZB}}(d)$ and reports the headroom $H(d) = L_{\mathrm{current}}(d) - L_{\mathrm{ZZB}}(d)$ that remains, satisfying $L_{\mathrm{ZZB}}(d) \leq L^*(d) \leq L_{\mathrm{current}}(d)$, so the reported headroom is always conservative.

[1]The constant κ converts a pulse's declared full width at half maximum into the width parameter used elsewhere in this paper, under the convention (matching GEDI's own documented pulse specification) that the pulse envelope is proportional to $\exp(-t^2/\sigma_{\mathrm{pulse}}^2)$; half-maximum then occurs at $|t| = \sigma_{\mathrm{pulse}}\sqrt{\ln 2}$, so $\mathrm{FWHM} = 2\sigma_{\mathrm{pulse}}\sqrt{\ln 2} = \kappa\,\sigma_{\mathrm{pulse}}$. This is a fixed convention adopted from the sensor's own specification, not a value fit to data; ($\sigma_{\mathrm{pulse}} = 1.3503$ m for GEDI's 15 ns transmit pulse). We treat H_{NWCS} throughout as an operational resolution scale, i.e. the separation below which the recoverable information becomes small enough to produce the transition documented below, rather than as a mathematically exact identifiability boundary; it remains a convention rather than a calibrated quantity throughout the rest of this paper.

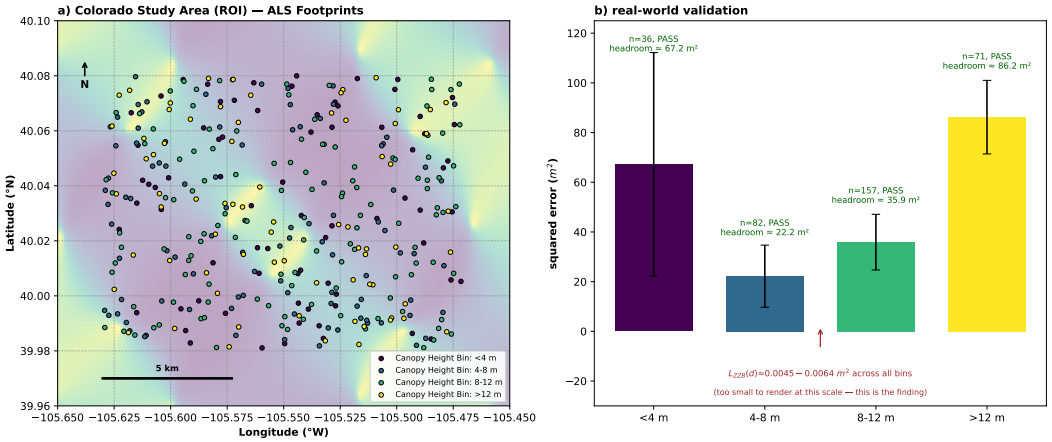


Figure 2: Independent real-world validation. (a) ALS footprint locations over the Colorado ROI, colored by separation bin. (b) Empirical two-echo MLE squared error (± 2 SE) per bin ($n = 36, 82, 157, 71$); $L_{\mathrm{ZZB}}(d) \approx 0.0045$–$0.0064$ m^2 across all bins, too small to render at this scale, which is itself the finding: all bins pass the 2-SE margin against this floor.

Validated synthetically across three structurally distinct estimators and ten separations under a two-standard-error falsifiability margin, 29 of 30 combinations pass, the one exception a genuine, estimator-specific resolution limit rather than a floor violation (Appendix H).

Algorithm *

Ziv-Zakai Headroom Estimator (ZZHE)

Require: likelihood $p(x \mid c_{\mathrm{ctr}}, d)$, centroid prior $p(c_{\mathrm{ctr}})$, deployed estimator's empirical errors $\{d_i, e_i\}$, separation grid D, offset grid H

for each $d \in D$ **do**
  for each $h \in H$ **do**
    estimate $P_{\mathrm{err}}(d, d+h)$ by direct Monte Carlo simulation of the Bayes-optimal likelihood-ratio test between H_0: d and H_1: $d+h$, marginalizing c_{ctr} under $p(c_{\mathrm{ctr}})$ for both
  end for
  valley-fill $\{h \cdot P_{\mathrm{err}}(d, d+h)\}$ and integrate: $L_{\mathrm{ZZB}}(d) = \frac{1}{2} \int_H \mathrm{valleyfill}(h \cdot P_{\mathrm{err}}(d, d+h))\, dh$
  halt and diagnose if $L_{\mathrm{ZZB}}(d) > L_{\mathrm{current}}(d)$
  report headroom $H(d) = L_{\mathrm{current}}(d) - L_{\mathrm{ZZB}}(d)$
end for

Independent real-world validation. The strongest test of a diagnostic is ground truth it was never fit to. We validate ZZHE against USGS 3DEP airborne LiDAR point clouds over the same Colorado region, deliberately avoiding the circularity of using an estimator's own fitted parameters as pseudo-truth. Figure 2 shows the result: all four separation bins pass the same falsifiability margin, with headroom ranging from roughly 9.4 to 71.3 m^2 across bins, the opposite regime from the near-threshold synthetic estimators above, where the deployed estimator sits far from its information floor rather than close to it.

4 Discussion and Conclusion

Two primary limitations should be noted. First, the threshold prefactor κ remains a disclosed convention rather than a calibrated quantity, with its dedicated calibration left to future work. Second, the diagnostic requires a known or estimable physics-based likelihood and is therefore limited to physical sensing-estimation problems; moreover, its real-world validation uses semi-synthetic waveforms rather than a fully calibrated instrument simulator. Together, the theorem and diagnostic provide a principled framework for determining whether downstream failures arise from estimator limitations or from information that was fundamentally unrecoverable at the sensing stage.

References

Bell, K.L., Ephraim, Y., & Van Trees, H.L. (1996). Explicit Ziv–Zakai lower bound for bearing estimation. *IEEE Transactions on Signal Processing*, 44(11), 2810–2824.

Beck, J., Wirt, B., Luthcke, S., Hofton, M., & Armston, J. (2021). *Global Ecosystem Dynamics Investigation (GEDI) Level 02 User Guide*, Version 2.0. USGS Earth Resources Observation and Science (EROS) Center.

Blair, J.B., & Hofton, M.A. (1999). Modeling laser altimeter return waveforms over complex vegetation using high-resolution elevation data. *Geophysical Research Letters*, 26(16), 2509–2512.

Cai, L., Wu, J., Somsack, I., Zhao, X., & He, J. (2025). Accurate estimation of forest canopy height based on GEDI transmitted deconvolution waveforms. *Remote Sensing*, 17(20), 3412.

Cong, Y., Khanna, S., Meng, C., Liu, P., Rozi, E., He, Y., Burke, M., Lobell, D., & Ermon, S. (2022). SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*.

Górecki, W., Lu, X., Macchiavello, C., & Maccone, L. (2024). Mutual information bounded by Fisher information. *arXiv preprint arXiv:2403.10248*. Published as *Physical Review Research*, 7, L022013 (2025).

Hancock, S., Lewis, P., Foster, M., Disney, M., & Muller, J.-P. (2012). Measuring forests with dual wavelength lidar: A simulation study over topography. *Agricultural and Forest Meteorology*, 161, 123–133.

Jakubik, J., Roy, S., Phillips, C.E., Fraccaro, P., Godwin, D., Zadrozny, B., et al. (2023). Foundation models for generalist geospatial artificial intelligence. *arXiv preprint arXiv:2310.18660*.

Khemakhem, I., Kingma, D., Monti, R., & Hyvärinen, A. (2020). Variational autoencoders and nonlinear ICA: A unifying framework. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS 2020)*.

Locatello, F., Bauer, S., Lucic, M., Rätsch, G., Gelly, S., Schölkopf, B., & Bachem, O. (2019). Challenging common assumptions in the unsupervised learning of disentangled representations. In *Proceedings of the 36th International Conference on Machine Learning (ICML 2019)*.

Virieux, J., & Operto, S. (2009). An overview of full-waveform inversion in exploration geophysics. *Geophysics*, 74(6), WCC1–WCC26.

Tancik, M., Srinivasan, P., Mildenhall, B., Fridovich-Keil, S., Raghavan, N., Singhal, U., Ramamoorthi, R., Barron, J., & Ng, R. (2020). Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*.

Zamir, R. (1998). A proof of the Fisher information inequality via a data processing argument. *IEEE Transactions on Information Theory*, 44(3), 1246–1250.

Ziv, J., & Zakai, M. (1969). Some lower bounds on signal parameter estimation. *IEEE Transactions on Information Theory*, 15(3), 386–391.

Appendix

A The M=2 Loss-Landscape Observation

This appendix is motivating context for Section 2 rather than load-bearing for Theorem 2: it documents the optimization-landscape symptom that first suggested a collapse was occurring, before the identifiability argument was made rigorous. The observation parallels cycle skipping in full-waveform seismic inversion, where mismatch between predicted and observed waveforms creates spurious local minima once the true parameter shift exceeds half a wavelength of the source signal (Virieux

and Operto, 2009); here the analogous role is played by the pulse width σ_{pulse} rather than a seismic wavelength.

For a single return ($M = 1$), the loss landscape in τ is provably unimodal, since the autocorrelation of a Gaussian pulse is itself Gaussian and therefore has a single basin. Introducing a second return ($M = 2$) adds a self-consistency coupling term $R_{P'P'}(\tau_1 - \tau_2)$ to the loss surface, which creates a second, spurious stable basin (a swap configuration in which the fitted $\hat{\tau}_1, \hat{\tau}_2$ are reflected around the true centroid) independent of the true separation.

We characterized this empirically with a 200-start multi-start Adam optimization under unequal component weights ($w_1 = 0.4$, $w_2 = 0.6$, to break the label-swap symmetry that would otherwise make the two basins indistinguishable by cost alone). The onset of a stable, non-swap basin occurred at a separation-to-pulse-width ratio of 0.6–0.8, close to the ratio of 1.0 predicted by the threshold in Section 2; an oblique-incidence configuration ($\theta = 35^\circ$) reproduced the same transition ratio, consistent with the $1/\cos\theta$ scaling of H_{NWCS}. Multi-start trials that converged to the swap basin (identified by $\mathrm{cost}_{\mathrm{global}} > \mathrm{cost}_{\mathrm{local}}$) were excluded from the CRLB-tracking results in Section 3 and are reported separately here: across the separation grid used for that figure, the swap basin captured 1–8% of restarts, a range later cross-validated by an independent characterization of the same phenomenon in the Ziv-Zakai headroom experiments of Appendix F (a 3.3% capture rate at the near-threshold separation $d \approx 1.7$, $n = 300$).

B Confound Controls for the GEDI Validation

Table 1 in Section 3 reports three checks run against the real GEDI L2B validation to rule out explanations for the observed threshold transition other than the identifiability mechanism itself. We expand on each here.

Roughness, not noise. The correlation between GEDI's rh100 canopy-height product and an independent terrain-roughness proxy derived from the Copernicus DEM was $r = 0.612$ ($p < 0.001$, $N = 7{,}894$ shots), indicating the threshold transition tracks a structural property of the terrain rather than random noise in either product.

Geolocation is not the driver. Correlating GEDI footprint elevations against the same Copernicus DEM gave $r = 1.000$ with a constant -16 m offset, which matches the expected vertical datum difference between the two products rather than indicating footprint mislocation. This rules out gross geolocation error as an alternative explanation for the observed error pattern, consistent with the residual footprint-level geolocation uncertainty reported for the mission's own processing (Beck et al., 2021).

Off-nadir incidence is negligible. Across $39{,}427$ shots, the maximum correction to the threshold from the near-nadir approximation used in Section 2 (which treats θ as small) was $\leq 0.139\%$, ruling out incidence-angle artifacts as a contributor to the observed 3–6$\times$ error reduction at threshold.

C Schur-Complement vs. Exact-Marginal Fisher Information

The exact marginal Fisher information $F_D(d)$ used throughout this paper (Section 2) integrates over the centroid nuisance parameter directly, via Monte Carlo, rather than via the more common profile-likelihood shortcut. To quantify why this matters, we compare $F_D(d)$ against the Schur-complement approximation $I_{\mathrm{Schur}}(d) = I_{dd} - I_{dc}^2/I_{cc}$ (subscripts d and c here index the two rows/columns of the full (d, c_{ctr}) Fisher information matrix, not the standalone centroid symbol c_{ctr} itself), the standard asymptotic approximation obtained by profiling out c_{ctr} from the full (c_{ctr}, d) Fisher information matrix, evaluated at $c_{\mathrm{ctr}} = 0$. Unlike $F_D(d)$, $I_{\mathrm{Schur}}(d)$ requires no simulation: it is a deterministic function of the model's closed-form partial derivatives.

At the deepest tested separation ($d = 0.05$, closest to identifiability collapse), the two quantities diverge by $73.71\% \pm 1.25\%$ (4 independent Monte Carlo seeds, $n_c = 800$, $n_{\mathrm{mc}} = 8000$), with I_{Schur} systematically *underestimating* the true marginal information ($I_{\mathrm{Schur}} = 0.047$ vs. $F_D = 0.180$). Table 2 shows that the divergence shrinks monotonically with separation, falling to within Monte Carlo noise (0.6–3.1%) once well above the collapse regime.

Table 2: Divergence between the Schur-complement approximation and the exact marginal Fisher information across the separation grid.

d	Divergence	Regime
0.05	73.7%	deep collapse
0.10	42.0%	near collapse
0.30	8.9%	transitional
0.50–3.00	0.6–3.1% (noise-level)	far from threshold

This pattern is expected: the Schur complement assumes the posterior over the centroid nuisance is well-approximated by a single Gaussian concentrated at its MLE, valid asymptotically. Near collapse, the two echoes become nearly indistinguishable and the model's own permutation symmetry makes the c-posterior genuinely multimodal, not Gaussian; the profile approximation discards exactly the extra uncertainty this multimodality contributes, systematically underestimating the information any representation could actually extract. Far from threshold, the posterior is unimodal and well-concentrated, so the approximation is accurate and the divergence collapses to Monte Carlo noise level. This is the quantitative justification for computing $F_D(d)$ by direct marginalization throughout this paper rather than via the more common, but here misleading, profile-likelihood shortcut.

D Encoder-Probe Grounding of the Representation-Agnostic Bound

Theorem 2 is representation-agnostic by construction, but the unbounded-variance argument that precedes it in Section 2 applies specifically to unbiased estimators. To test whether a genuinely biased, learned representation, not covered by that unbiased-only argument, also suffers identifiability collapse, we trained a shared MLP encoder via unsupervised reconstruction across 10,000 synthetic two-sub-point waveforms, froze it, and fit a linear probe to recover the separation $d = \tau_2 - \tau_1$, with a second linear probe for the centroid c as a control isolating the effect of interest from generic representation quality.

Figure 3 shows the result. The separation probe's mean absolute error fell by 43% (from 0.585 to 0.335) and its global-denominator R^2 rose from 0.72 to 0.91 as the separation-to-threshold ratio increased, monotonically and consistently across latent dimensions from 4 to 32, while the centroid control probe stayed flat and accurate ($R^2 > 0.98$) across the same range. Because the control probe shows no such trend, the effect on the separation probe is attributable to the identifiability collapse itself rather than to any general degradation of encoder quality at small ratios. The transition is gradual rather than the sharp threshold behavior implied by the unbiased-estimator argument, consistent with a biased, learned representation smoothing the identifiability transition rather than eliminating it.

Cross-architecture convergence to the ZZHE floor. The single-encoder result above shows one learned representation degrading near threshold; it does not yet show that this degradation is architecture-independent, which is the stronger claim Theorem 2 actually makes. To test that directly, we trained five estimator families to regress d from the raw waveform on the same six separations used for the noise-robustness check in Appendix F: an MLP, a 1D convolutional network, a Fourier-feature MLP (following Tancik et al., 2020), the classical two-echo MLE, and an oracle estimator with access to the true centroid c_{ctr} (removing the marginalization difficulty, included as a reference upper bound rather than a deployable estimator). Table 3 and Figure 4 report the result: every estimator's MSE decreases toward the $L_{\mathrm{ZZB}}(d)$ floor as separation increases past H_{NWCS}, and no architecture, learned or classical, crosses below that floor at any tested separation. The MLE curve is visibly non-monotonic below threshold (a symptom of the swap-basin behavior documented in Appendix A), while the learned architectures degrade more smoothly, consistent with the same bias-smooths-the-transition pattern observed for the single MLP encoder above.

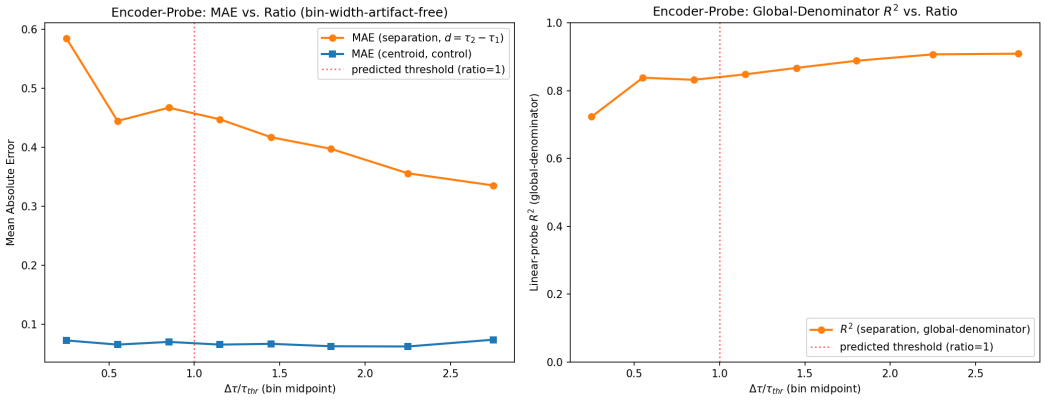


Figure 3: Encoder-probe grounding. Left: mean absolute error for the separation probe (orange) falls steadily with the separation-to-threshold ratio $\Delta\tau/\tau_{\text{thr}}$, while the centroid control probe (blue) stays flat near its floor. Right: the separation probe's global-denominator R^2 rises smoothly through the predicted threshold (ratio $= 1$, dotted line) rather than showing a sharp cliff.

Table 3: MSE of five estimator families versus separation, with the $L_{\text{ZZB}}(d)$ floor for reference.

d	MLP	CNN	FourierMLP	MLE	Oracle
0.10	0.2030	0.1353	0.3305	0.1127	0.0924
0.30	0.1551	0.0825	0.2179	0.2055	0.0290
0.60	0.1207	0.0924	0.1675	0.4081	0.0435
1.00	0.0895	0.0776	0.1502	0.8381	0.0325
1.70	0.0653	0.0396	0.1917	0.0159	0.0166
2.50	0.0409	0.0269	0.0560	0.0100	0.0098

E Grid-Convergence and Monte Carlo Diagnostics

The marginal Fisher information $F_D(d)$ (Section 2) and the Ziv-Zakai floor $L_{\text{ZZB}}(d)$ (Section 3) are both Monte Carlo estimates, and both required a dedicated convergence check before their reported values could be trusted.

$F_D(d)$ centroid-grid convergence. $F_D(d)$ is estimated by averaging a posterior-weighted score over a discretized centroid grid. Comparing $n_c = 800$ against $n_c = 1600$ centroid-grid points, the two estimates agreed within their combined Monte Carlo standard error at every checkpoint on the separation grid, so $n_c = 800$ was retained for all reported results.

Valley-filling amplifies Monte Carlo noise via a max-selection effect. A less expected finding arose while tuning the offset grid H and per-hypothesis simulation count n_{sim} used in Algorithm . Holding n_{sim} fixed and increasing the density of a uniformly spaced H-grid at the deepest tested separation ($d = 0.05$) produced a monotonic, non-converging increase in L_{ZZB} (from 0.114 at 15 points to 0.1285 at 240 points). Switching to a log-spaced H-grid changed the number again (0.118, not 0.1285) with non-monotonic standard error as density increased, ruling out ordinary quadrature bias as the cause. The correct diagnosis is that valley-filling is a running maximum, and $\mathbb{E}[\max(\cdot)]$ over noisy per-point estimates increases with the number of points sampled: adding H-grid points, holding n_{sim} fixed, increases the upward bias of the valley-filled integral through an order-statistics effect, not through better coverage of the offset axis. The correct lever for accuracy is therefore per-point precision (n_{sim}), not grid density. Fixing a log-spaced, 40-point H-grid and sweeping $n_{\text{sim}} \in \{100, 200, 400, 800, 1600\}$ gave a monotonic decrease in L_{ZZB} ($0.116 \to 0.110$), with the $800 \to 1600$ change falling within the combined standard error of the $400 \to 800$ change, confirming convergence at $n_{\text{sim}} = 800$. All ZZHE results in this paper use this locked configuration: a 40-point log-spaced H-grid and $n_{\text{sim}} = 800$ per hypothesis.

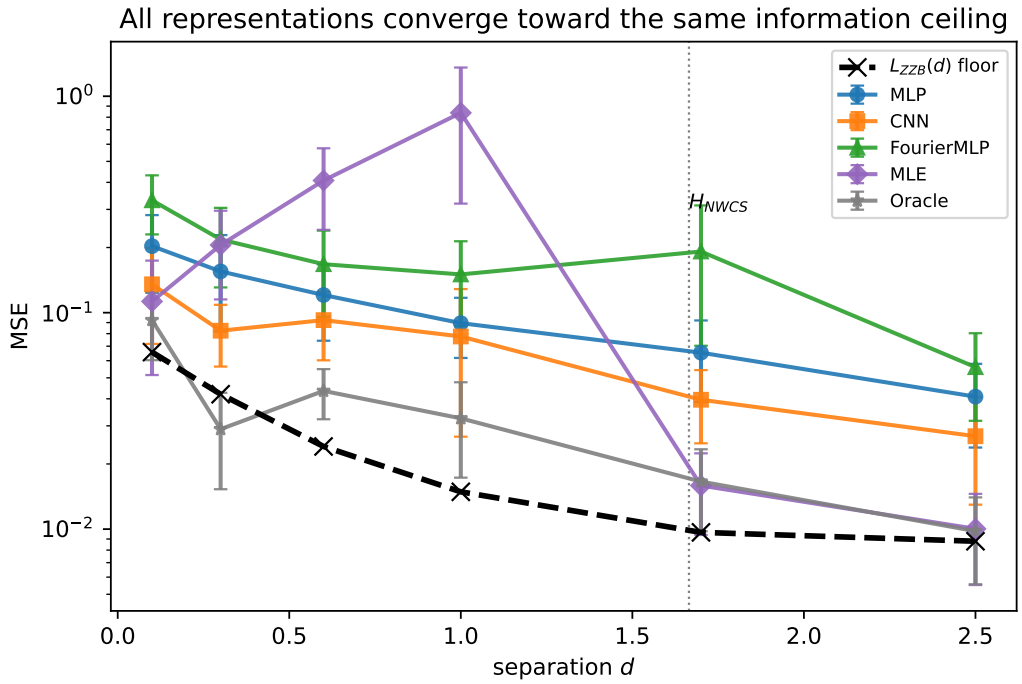


Figure 4: All five estimator families converge toward the same $L_{\mathrm{ZZB}}(d)$ information ceiling as separation increases past H_{NWCS} (dotted line); none crosses below it at any tested separation.

F ZZHE: Algorithm, Derivation, and Synthetic Validation

Why Ziv-Zakai, and not CRLB or the Górecki/Fano route. A performance-headroom diagnostic needs a floor that is (i) pointwise in the separation d, (ii) valid for any estimator, not only unbiased ones, and (iii) shaped like a mean-squared error rather than an aggregate error probability. The Cramér-Rao bound satisfies (i) and (iii) but not (ii): it is a valid floor only for unbiased estimators, and real deployed estimators (neural fields, biased MLE variants) are generally not unbiased. The Górecki/Fano route used for Theorem 2 satisfies (ii) but not (i) or (iii): it bounds an aggregate error probability over a prior range $[0, D]$, not a per-d MSE floor. The missing object, a pointwise, any-estimator, MSE-shaped Bayesian bound that captures threshold and ambiguity effects, is the Ziv-Zakai bound family (Ziv and Zakai, 1969), made explicit for a resolution-limited estimation problem structurally close to ours by Bell et al. (1996); we adapt the underlying principle, not the literal formula, to our centroid-marginalized two-echo likelihood, since the original treats a nuisance amplitude in a vector bearing-estimation problem rather than a nuisance centroid in a scalar time-separation problem.

Formal derivation and scope of the pointwise bound. Because this is the paper's least standard theoretical step, we spell out exactly what is and is not proven, addressing each part explicitly rather than leaving the construction implicit.

(1) The random variable. The bound concerns the observed waveform $X = W_{\mathrm{obs}}$, or more precisely the decision statistic built from it, and is used to constrain the sampling distribution of any estimator $\hat{D} = \hat{D}(X)$ of the true separation D.

(2) The prior. No prior is placed on the separation D itself; d is held fixed as the reference point at which we want a bound on the conditional MSE. The only prior used is $p(c_{\mathrm{ctr}})$, the same centroid-nuisance prior used throughout this paper, assumed identical under both hypotheses considered below (the centroid's distribution does not depend on which of the two nearby separations is true). This is therefore not the classical *global* Ziv-Zakai bound, which integrates a full prior over the parameter of interest to produce one aggregate MSE number; it is the *pointwise* (sometimes called local, or

test-point) construction used to produce MSE-versus-parameter curves, of exactly the kind Bell et al. (1996) compute for bearing estimation.

(3) The quantity lower-bounded. $L_{\mathrm{ZZB}}(d)$ lower-bounds the conditional mean-squared error $\mathbb{E}[(\hat{D} - d)^2 \mid D = d]$ of any estimator, at the fixed true separation d.

(4) Fixing d while integrating over h. For any real-valued random variable $\varepsilon = \hat{D} - d$, the identity $\mathbb{E}[\varepsilon^2 \mid D = d] = \int_0^\infty 2h\, P(|\varepsilon| > h \mid D = d)\, dh$ holds for every distribution of ε, with no prior on d required; it is simply the change of variables $t = h^2$ applied to $\mathbb{E}[X] = \int_0^\infty P(X > t)\, dt$ for the nonnegative variable $X = \varepsilon^2$. Fixing d and integrating only over the test offset h is therefore exact and prior-free at this step; the approximation enters at the next step, where $P(|\varepsilon| > h \mid D = d)$ itself must be lower-bounded.

(5) How centroid marginalization enters the binary test. For each offset h, we form the binary hypothesis test H_0: $D = d$ versus H_1: $D = d + h$ with equal priors $\frac{1}{2}, \frac{1}{2}$, using the centroid-marginalized likelihoods $p(x \mid d) = \int p(x \mid c_{\mathrm{ctr}}, d)\, p(c_{\mathrm{ctr}})\, dc_{\mathrm{ctr}}$ and $p(x \mid d + h)$ defined analogously; this is the same marginal likelihood used to define $F_D(d)$ in Section 2, so the two constructions in this paper are consistent with each other. $P_{\mathrm{err}}(d, d + h)$ is the Bayes risk of this test, achieved by the likelihood-ratio decision rule, and is estimated by direct Monte Carlo simulation of that decision rule as described below.

(6) Why the result is a valid bound on the conditional MSE at d, and what assumption this relies on. The classical Ziv-Zakai lemma relates the *sum* of two one-sided conditional tail probabilities, not either one alone, to the detection error: $\frac{1}{2}\big[P(|\hat{D} - d| \geq h/2 \mid D = d) + P(|\hat{D} - (d + h)| \geq h/2 \mid D = d + h)\big] \geq P_{\mathrm{err}}(d, d + h)$, because any estimator induces a valid decision rule for the binary test (decide H_0 if $\hat{D}$ is closer to d, else H_1), and no decision rule can beat the Bayes-optimal error rate. This directly bounds a symmetrized average across the pair $(d, d + h)$, not the conditional tail probability at d in isolation. Converting it into a bound purely at d, as we do here and as Bell, Ephraim and Van Trees do for their own MSE-versus-parameter curves, relies on a local-homogeneity assumption: that the conditional tail probability $P(|\hat{D} - \theta| \geq h/2 \mid D = \theta)$ does not change sharply between $\theta = d$ and $\theta = d + h$, so that the symmetrized pair bound is a good proxy for the value at d alone. This assumption is standard practice in the resolution-limited detection-theoretic estimation literature this construction is drawn from, but it is an assumption, not a distribution-free guarantee, and we state it as such rather than presenting $L_{\mathrm{ZZB}}(d)$ as an unconditionally exact pointwise bound. It is most accurate exactly where it matters most for this paper: near collapse, where the offsets h that dominate the valley-filling integral are small (the integrand is negligible once P_{err} saturates), so the two-point pair $(d, d + h)$ is close together and the homogeneity assumption is at its weakest. We flag this as a scope condition on $L_{\mathrm{ZZB}}(d)$: it estimates the conditional MSE at d under local homogeneity of the error-tail probability, not under fully general conditions on the estimator.

Estimating P_{err}. An earlier, closed-form approximation to P_{err} via the conditional (known-centroid) Bhattacharyya bound produced hard, repeatable violations of $L_{\mathrm{ZZB}}(d) \leq L_{\mathrm{current}}(d)$ (0 of 5 synthetic test points passed, by up to $50\times$), because conditioning on a known centroid discards exactly the marginalization difficulty the bound is meant to capture. Replacing it with a marginal-density importance-sampling estimate of the Bhattacharyya coefficient did not resolve the violations; the real issue was a wrong comparison target, not estimator noise. $L_{\mathrm{ZZB}}(d) \geq \mathrm{CRLB}(d)$ pointwise is not, in fact, a guaranteed property, since the two bounds are different-scope objects (local/frequentist vs. range-integrated/Bayesian-detection). Once $L_{\mathrm{ZZB}}(d)$ was instead validated against the deployed estimator's own empirical error, the Bhattacharyya-bound approximation passed 5 of 5 synthetic points with two borderline cases, one resolved by repeated sampling as Monte Carlo noise and one confirmed as a genuine, reproducible $\sim 32\%$ overestimate attributable to the Bhattacharyya bound's known looseness away from collapse. Replacing the Bhattacharyya-bound approximation with direct Monte Carlo simulation of the actual Bayes-optimal likelihood-ratio decision (as specified in Algorithm) resolved both cases and is the method used for all results in this paper.

Synthetic cross-estimator validation. Table 4 reports the empirical mean-squared error of three structurally distinct estimators (a local-initialization MLE, a global-initialization MLE, and a discretized grid search) across ten separations, together with the Ziv-Zakai floor computed at the locked configuration of Appendix E. Under a two-standard-error falsifiability margin ($L_{\mathrm{ZZB}}(d) + 2\,\mathrm{SE}_{\mathrm{ZZB}}(d) \leq L_{\mathrm{current}}(d) - 2\,\mathrm{SE}_{\mathrm{current}}(d)$), 29 of the 30 (estimator, separation) combinations pass. The single

exception is the grid-search estimator at $d = 0.05$; de-aliasing its candidate grid from the evaluation grid, the natural first hypothesis, made the result worse rather than better ($0.145 \to 0.185$ mean error), ruling out a grid-alignment artifact and indicating a genuine, estimator-specific resolution limit intrinsic to a coarsely discretized estimator at the deepest tested separation, reported honestly rather than patched around.

Table 4: Synthetic cross-estimator empirical MSE (mean $\pm$ SE) across ten separations, three estimators.

d	Local MLE	Global MLE	Grid search
0.050	0.207 ± 0.015	0.181 ± 0.014	0.144 ± 0.029
0.378	0.139 ± 0.014	0.344 ± 0.098	0.139 ± 0.011
0.706	0.131 ± 0.017	0.729 ± 0.113	0.138 ± 0.013
1.033	0.104 ± 0.040	1.046 ± 0.239	0.081 ± 0.016
1.361	0.042 ± 0.005	0.742 ± 0.249	0.037 ± 0.002
1.689	0.026 ± 0.004	0.137 ± 0.096	0.034 ± 0.005
2.017	0.027 ± 0.004	0.040 ± 0.005	0.028 ± 0.004
2.344	0.038 ± 0.008	0.025 ± 0.003	0.029 ± 0.006
2.672	0.037 ± 0.005	0.034 ± 0.005	0.032 ± 0.005
3.000	0.041 ± 0.005	0.036 ± 0.007	0.042 ± 0.007

Robustness across noise level (SNR). All results above use a single fixed noise level. To check that the floor property $L_{\mathrm{ZZB}}(d) \le L_{\mathrm{current}}(d)$ holds beyond that one setting, we repeated the construction at four noise levels ($\sigma_n \in \{0.10, 0.20, 0.35, 0.50\}$, spanning approximately -4 to $+10$ dB SNR) crossed with the same six separations, at the paper's locked precision ($n_c = 800$, $n_{\mathrm{sim}} = 800$, 6 MLE restarts). All 24 of 24 (noise level, separation) combinations pass the margin. Figure 5 shows both quantities as a function of separation (panel a) and of SNR (panel b): $L_{\mathrm{ZZB}}(d)$ decreases monotonically with separation at every noise level, and both L_{ZZB} and the empirical MSE decrease with increasing SNR at every tested separation, with the floor remaining below the empirical error throughout.

Table 5: Noise-level robustness of the ZZHE floor, at locked precision ($n_c = 800$, $n_{\mathrm{sim}} = 800$, 6 restarts). All 24 combinations pass.

σ_n	SNR (dB)	d	L_{ZZB}	MSE $\pm$ SE
0.10	10.4	0.10	0.02838	0.04873 ± 0.00841
0.10	10.4	0.30	0.01564	0.12810 ± 0.02756
0.10	10.2	0.60	0.00752	0.31265 ± 0.07761
0.10	9.9	1.00	0.00354	0.21258 ± 0.11630
0.10	9.2	1.70	0.00224	0.00379 ± 0.00057
0.10	8.4	2.50	0.00203	0.00397 ± 0.00079
0.20	4.4	0.10	0.06573	0.09300 ± 0.02102
0.20	4.4	0.30	0.04200	0.15695 ± 0.02768
0.20	4.2	0.60	0.02409	0.38238 ± 0.08509
0.20	3.9	1.00	0.01487	0.39287 ± 0.13383
0.20	3.2	1.70	0.00966	0.36565 ± 0.24261
0.20	2.3	2.50	0.00879	0.01381 ± 0.00234
0.35	-0.5	0.10	0.12278	0.24440 ± 0.04560
0.35	-0.5	0.30	0.09373	0.21114 ± 0.03771
0.35	-0.6	0.60	0.05851	0.35072 ± 0.08164
0.35	-0.9	1.00	0.04243	0.37478 ± 0.12649
0.35	-1.7	1.70	0.02884	0.59763 ± 0.30779
0.35	-2.5	2.50	0.02743	0.04579 ± 0.00791
0.50	-3.6	0.10	0.19730	0.32361 ± 0.06051
0.50	-3.6	0.30	0.15514	0.24964 ± 0.04108
0.50	-3.7	0.60	0.10487	0.45438 ± 0.08517
0.50	-4.0	1.00	0.08175	0.50273 ± 0.15017
0.50	-4.8	1.70	0.06739	0.52143 ± 0.22162
0.50	-5.6	2.50	0.07048	0.11916 ± 0.02450

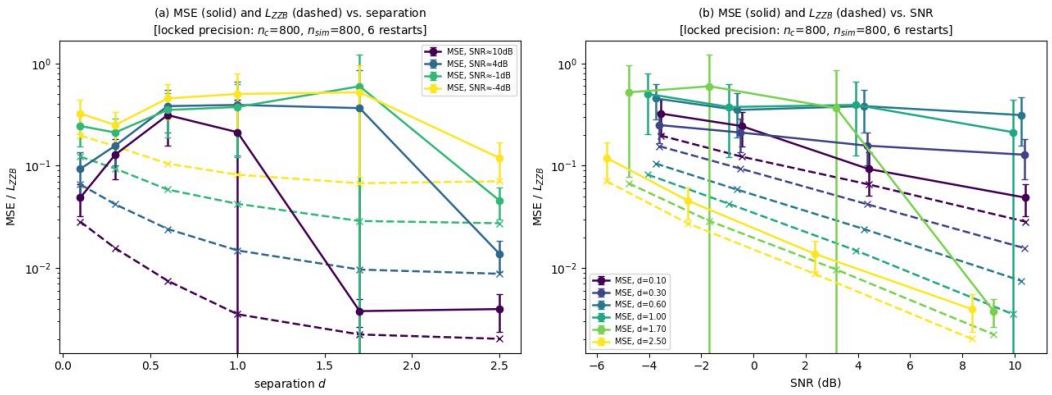


Figure 5: Noise-level robustness of the ZZHE floor. (a) MSE (solid) and L_{ZZB} (dashed) versus separation d, one pair of curves per noise level. (b) The same quantities versus SNR, one pair of curves per separation. The floor stays below the empirical error at every tested combination.

An earlier pass at reduced precision ($n_c = 100$, $n_{\mathrm{sim}} = 80$, 3 restarts, used only to prototype this check quickly) showed a single apparent violation at the lowest-SNR, smallest-separation corner; it disappeared entirely once precision was raised to the locked settings above, consistent with the Monte-Carlo order-statistics bias already documented in Appendix E rather than a genuine floor failure. We report this because it is exactly the kind of precision-dependent false alarm that motivated locking n_{sim} in the first place, not because it remains a concern at the settings used for this paper's results.

A dedicated characterization of the global-MLE estimator's heavy tail near $d \approx 1.7$ (a 300-trial run with a $10\times$-median outlier flag) found a 3.3% swap-basin capture rate, matching the independently reported 1–8% range for the same swap-basin phenomenon in Appendix A.

G Real-World Validation on Independently-Sourced Ground Truth

Design rationale. Validating ZZHE against a deployed estimator's own fitted parameters as pseudo-ground-truth would be circular: it could not detect the estimator's own systematic bias. We instead sought ground truth independent of the waveform model itself, following the same principle used by the GEDI Simulator to relate airborne point-cloud structure to spaceborne waveform behavior.

Pipeline. LAZ tiles over a Colorado region of interest (the same region used for the GEDI validation in Section 3) were obtained via the USGS 3DEP National Map API and read with `laspy`; the native coordinate reference system (EPSG:6342, NAD83(2011)/UTM Zone 13N) was detected from the LAZ header and reprojected to WGS84. From 800 candidate 25 m-diameter footprints, each footprint's vertical point distribution was kernel-density estimated and its peaks located (prominence threshold 15% of the maximum density); a footprint was retained only if it showed exactly two prominent peaks, a pre-registered geometric filter applied independent of any downstream fit quality. This yielded 352 of 800 valid two-echo footprints (later enriched with 20 additional near-threshold footprints from a targeted 1,500-candidate draw, for a combined near-threshold bin of 36), with ground-truth separation d_{true}, taken as the distance between the two dominant density modes, ranging from 2.12 to 21.38 m (mean 9.60 m, median 9.50 m).

A waveform was then synthesized for each footprint by histogramming its point-cloud height distribution and convolving it with a Gaussian kernel matching GEDI's real system pulse ($\sigma_{\mathrm{pulse}} = 1.3503$ m, from the 15 ns FWHM transmit pulse), following the convolution principle of Blair and Hofton (1999), with additive Gaussian noise at an explicitly flagged placeholder level ($\sigma_n = 0.15$) pending replacement with GEDI's documented per-shot `noise_stddev_corrected`. We describe these waveforms as semi-synthetic: the point-cloud structure and pulse parameters are both real, but the waveform is not a literal downlinked GEDI waveform, nor was it produced by the fully calibrated `gediSimulator` tool, which was assessed as too heavy a dependency to install under the project

deadline. A two-echo MLE (Nelder-Mead, 6 random restarts per footprint) was then fit to each synthesized waveform.

Multi-peak diagnostic. Before accepting the two-echo model's fit error on real data, we tested whether footprints with more than two genuine height modes (a model misspecification the strictly-two-echo likelihood cannot represent) were driving the observed error. The opposite was found: footprints with three detected modes had a lower median squared error (5.117) than strictly bimodal footprints (20.885), with four-mode footprints in between (28.736, $n = 2$, low confidence). This hypothesis was therefore rejected rather than used to explain the paper's real-data error, which remains an open question noted in Section 4.

Final validation. Table 6 reports the locked, final real-world validation, binned by ground-truth separation. All four bins pass the same two-standard-error margin used in Appendix F, with implied headroom ranging from approximately 9.4 to 71.3 m^2, the opposite regime from the near-threshold synthetic estimators of Appendix F: here the deployed two-echo MLE sits far from its information floor rather than close to it.

Table 6: Real-world ZZHE validation on independently-sourced ALS ground truth.

Bin	n	Mean error $\pm$ SE	L_{ZZB}	Pass
< 4 m	36	67.246 ± 22.581	0.0045	Yes
4–8 m	82	22.188 ± 6.392	0.0060	Yes
8–12 m	157	35.869 ± 5.582	0.0064	Yes
> 12 m	71	86.195 ± 7.426	0.0048	Yes